Gene Essentiality Gate

The Held-Out Score, the Module Form That Carries, and the Panels Proposed for the Main Text

Michael Volk

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1 The decision ✗

Gene essentiality is to be its own main-text figure, separated from fitness and interactions, because the study behind it is closed on its two goals and its figure can carry a claim the others cannot yet. Every model number here is on the 195 genes of the 223 that Merzbacher 2025 held out of its own training and that resolve as single-deletion genotypes in the build, 31 of them essential, with the epoch chosen on validation folds that never contain them. Nothing here is approved. The sheets are the summary figures of `028-gene-essentiality`, and the symbol after each panel letter says only whether the values behind it are on file.

Where the full record is. `028-gene-essentiality` holds the study: the endpoint, its labels and splits; the comparators and how Flux Cone Learning's published numbers were reproduced (five-fold mean 0.780 published, 0.777 recomputed); the backbone rounds; every form of the metabolic module and why the layer was retired; the two model families and their average; the hard genes; and the summary on one page. This document takes its result files and asks what a figure can claim.

Three statements are on file. First, the held-out score. The best method is an equal-weight average of two families, GO convolution, five folds (400 epochs) with + GPR redundancy counts, pathway + ProtT5, five folds (round 80), at 0.955 AUROC [0.922, 0.982] on the 195 genes; the best single model reads 0.930 [0.886, 0.966], against 0.742 for Flux Cone Learning and 0.640 [0.519, 0.755] for FBA on the same genes. On 10 other held-out fifths of the labeled genes, with the average fixed in advance, it beats the better single model on 9 of 10 (+0.023). Second, the module form that carries. A single-medium parsimonious solve per genotype, run once offline and read as an affine on the logit beside six gene-rule counts, lifts the same cell with no metabolism by +0.061 on the 10 repeated splits, higher on 10 of 10 splits and on 89 of 90 split-and-seed pairs, and it matches the four-media form (+0.002, 6 of 10 splits). Every form that kept a linear program inside the training loop scored lower, so the layer is retired and the solve is kept as data. Third, the limit. Off the metabolic genes every deletion reads one constant row, and on the 4,674 genome-wide genes that carry no GEM reaction (974 essential) the transformer with the solve reads 0.570 [0.550, 0.590] where graph convolution on ProtT5 over the union of nine gene graphs reads 0.755 [0.737, 0.773].

The regime question, settled by naming it. The highest single cell in the study is transductive: the one-hop propagation of the training and validation genes' calls through the gene graphs. It is not on the proposed main-text sheet, because its number holds only inside a called screen. Every bar proposed for the main text is inductive, reading the gene and the metabolic model and nothing measured on the screen's other genes, and the sheet's key says so. Regime is what a method reads when it scores a held-out gene: *inductive*, the gene's own sequence, annotation, graph position and the metabolic model's solve for its deletion, none of which needs the screen; *transductive*, a measured call or phenotype of other genes in the same screen (the label-propagation store), which holds only inside a called screen; *comparator*, a published method scored on the same genes with nothing learned here.

The claims the main-text figure would make. First, that on the released held-out genes the method beats the published Flux Cone Learning and FBA by a margin the gene bootstrap separates: on file. Second, that the metabolic model enters best as data, one offline solve per genotype and medium read through an affine, and that this form carries across the two model families: on file. Third, that the metabolic module covers only the genes with a reaction and the gene graphs cover the rest, so a model over the whole database needs both inputs: on file. Fourth, that the hard genes share an annotation (isozyme redundancy): on file in `028-gene-essentiality`, not yet read into this document's numbers. The figure enters the paper when the composition of 2 is drawn at Nature size from the same result files and the author approves each claim's sentence.

2 The mockups ✗

Result files are under `experiments/028-gene-essentiality/results/`. The two sheets below are the summary figures of `028-gene-essentiality` as they stand; the proposed main-text figure is a composition of their panels, listed first, and is not drawn yet. The statistic is the AUROC on the 195 held-out genes under the fold protocol (five models trained on five cross-validation files, logits averaged on the held-out genes) unless the caption says otherwise; intervals are gene bootstraps.

The sheet proposed for the main text. Six panels in two rows: the setup and the score, then the module.

a The two families and their average, drawn as in the families explainer of `028-gene-essentiality`: the cell graph transformer with the offline solve on an affine, graph convolution on the GO multi-hot, and the standardized equal-weight average with nothing fitted. ●

b Every method on the 195 held-out genes, best first, with Flux Cone Learning and FBA as comparators: Figure 1a. ●

c The three fixed in advance on 10 other held-out fifths, three seeds per split: Figure 1b. ●

d The module forms on one backbone, seed means with every seed drawn, the in-loop forms hatched: Figure 2a. ●

e Each form minus the same cell with no metabolism on the 10 splits, and the share of split-and-seed pairs above: Figure 2e and f. ●

f The genome-wide 4,674 genes with no GEM reaction, where the solve reads a constant row and the gene graphs carry the signal: Figure 1c. ●

The hard-genes panel (Figure 1d and e) is proposed for the supplement with the misclassification enrichment table, and the cost panels (Figure 2b and c) for the Methods.

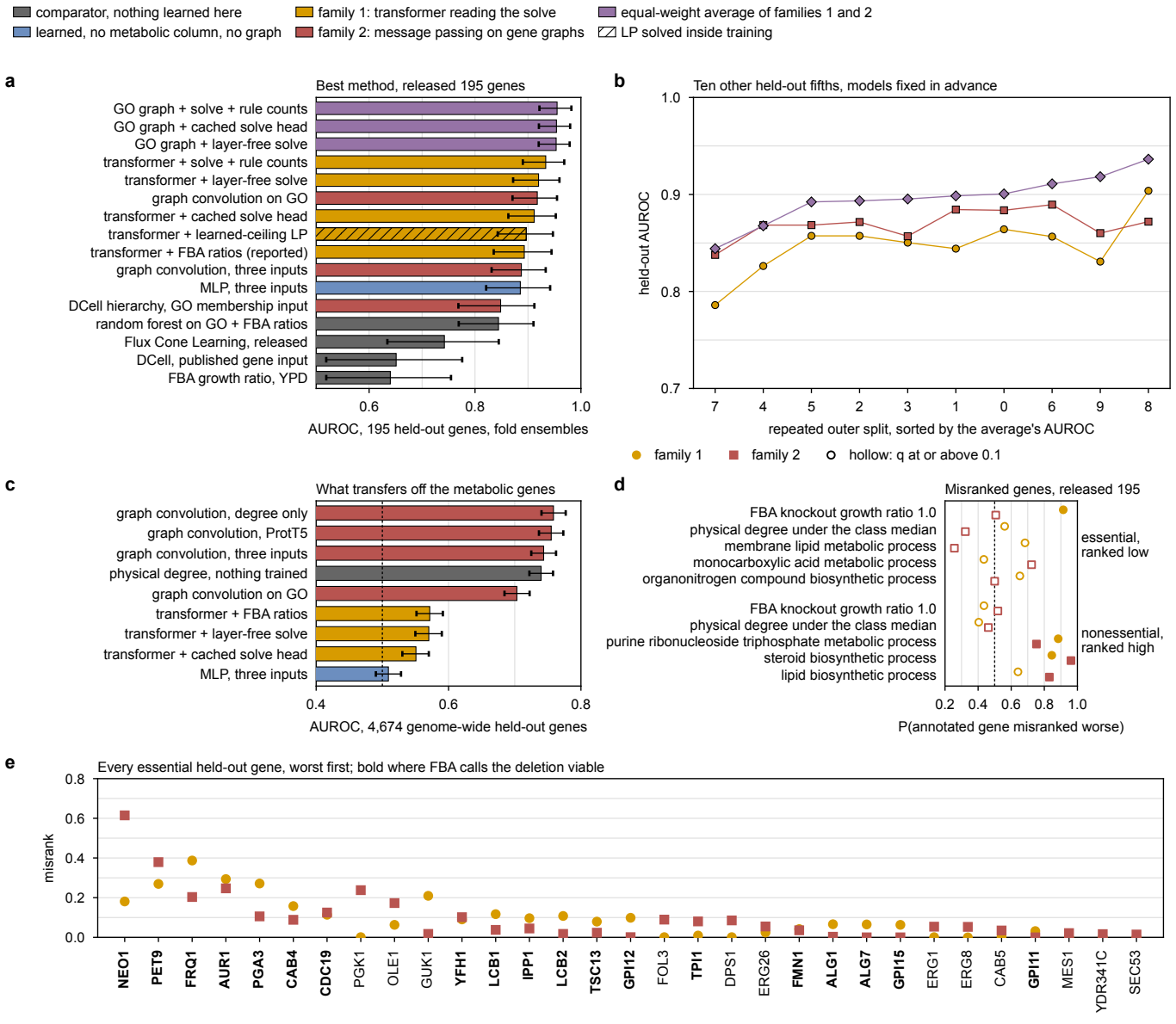


Figure 1: Goal 1: the best method on gene essentiality, against Flux Cone Learning and FBA. One color means one thing in every panel: gray a comparator with nothing learned here, blue a learned model reading no metabolic column and no graph, orange family 1 (the transformer), red family 2 (graph convolution on GO), purple their average; a hatch marks a linear program solved inside the training loop. **a**, Every method on the 195 held-out genes, best first; the first rows are averages of family 2 with a form of family 1 (the best at 0.955), the DCell-style rows are the comparator under the same fold protocol (0.848 [0.769, 0.912]), Flux Cone Learning reads 0.742 and FBA 0.640. Learned models are five-fold ensembles, the random forest is one fit on the split file, and intervals are gene bootstraps. **b**, The layer-free form of family 1 (circles), graph convolution on GO (squares) and their average (diamonds), fixed in advance, on 10 other held-out fifths of the labeled genes, three seeds per split, sorted by the average's AUROC; means 0.848, 0.869 and 0.896. **c**, The same models trained on the released genes and read on the 4,674 genome-wide genes that carry no GEM reaction, pooled over seeds with a gene bootstrap; the dashed line is chance. **d**, What the misranked genes of the released 195 share, per class, for the two families; filled where the Benjamini-Hochberg q within the family is under 0.1. **e**, Every one of the 31 essential genes in the released held-out set, worst first by the two families' mean misrank. Written by `summary_figure.py`. ●

What is not drawn. The families explainer is a draw.io composition in `028-gene-essentiality` and enters the proposed sheet as panel a once it is re-exported into this document's figures. The main-text composition itself, at Nature's full-page box, is the open item of 4.

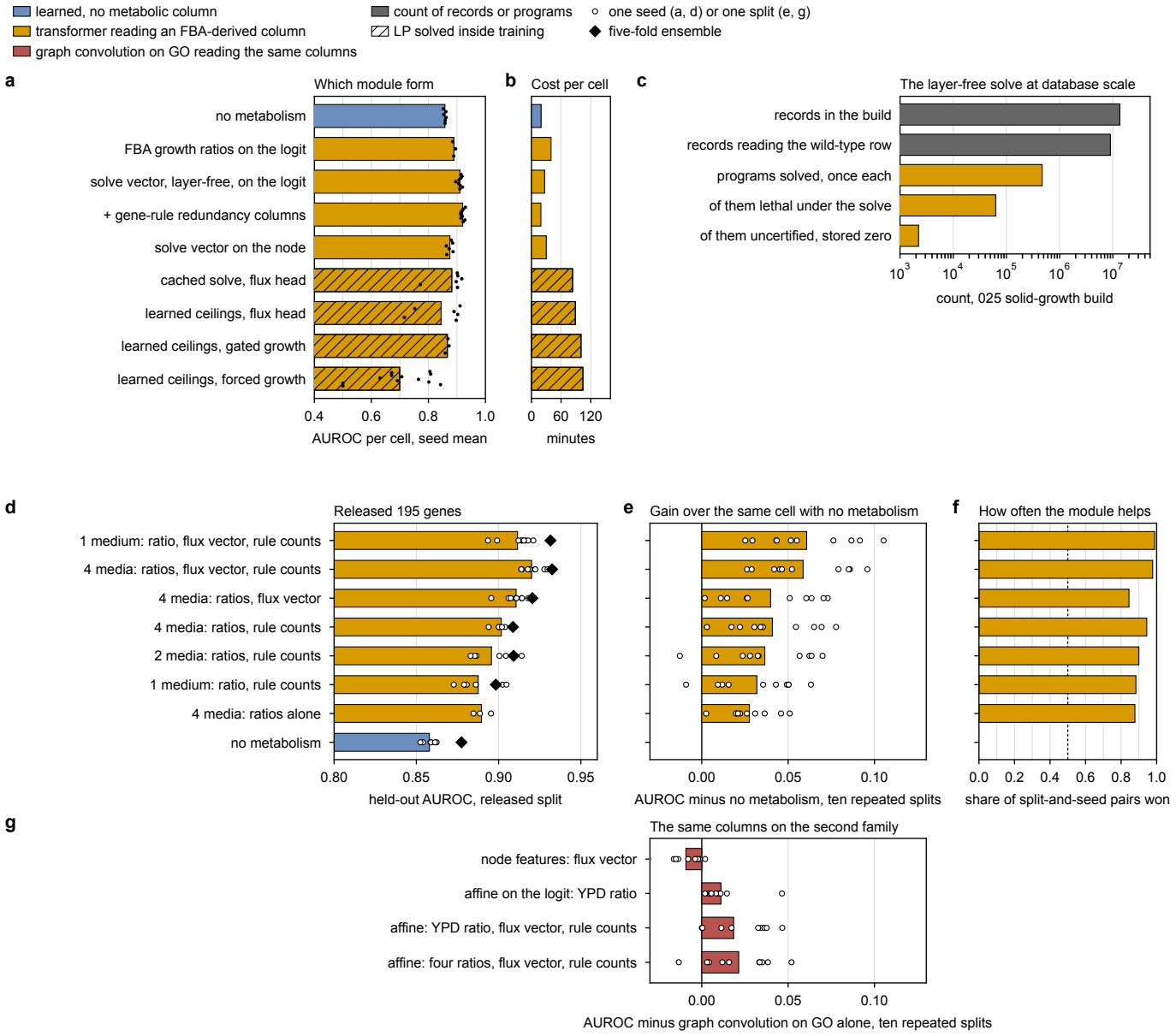


Figure 2: Goal 2: the form of the metabolic module that helps, what it costs, and what a record of one genotype in one medium can carry. Orange is the transformer reading any FBA-derived column, blue the same cell with no metabolic column, gray a count; a hatch marks a linear program solved inside the training loop. **a**, The forms of the module on one backbone, the pathway multi-hot with ProtT5: the bar is the seed mean of the held-out AUROC and each point one seed; no metabolism 0.858 (6 seeds, sd 0.004), the single-medium solve with the rule counts 0.912 (9 seeds, sd 0.009), the four-media form 0.920. **b**, Wall minutes per cell for the forms of **a**, as run. **c**, What the layer-free solve costs over the solid-growth build, logarithmic. **d**, The layer-free forms by the number of media whose growth ratio they read and by whether they also read the stored flux vector and the six gene-rule counts, on the released split; the diamond is the five-fold ensemble (0.932 for the single medium, 0.934 for four). **e**, Each form minus the same cell with no metabolism on the 10 repeated splits: the single-medium form reads +0.061, above on 10 of 10. **f**, The share of split-and-seed pairs on which the form scores above the cell with no metabolism (89 of 90 for the single-medium form); the dashed line is a module that helps on half the runs. **g**, The same columns handed to the second family, graph convolution on GO, on the repeated splits: as node features and as a zero-initialized affine on the node's logit, each minus the same convolution on GO alone. Written by `summary_figure.py` from `simplification_readout.json` and `resplit_readout.json`. ●

3 The models and comparators on file ✗

Family 1, the transformer. The cell graph transformer over the nine-graph perturbed gene graph with the pathway multi-hot and ProtT5 as node features, the deleted gene's token pooled to a logit, and a zero-initialized affine that adds the metabolic columns computed offline from yeast-GEM 9.0.2: the growth ratio of the deletion on one medium, the deletion's parsimonious flux vector as $\log(1 + |v|)$, and six counts read off the gene-protein-reaction rules. As a five-fold ensemble on the released split the single-medium form reads 0.929 [0.885, 0.966], the rule-count form 0.930 [0.886, 0.966], and the layer-free solve alone 0.920 [0.873, 0.957].

Family 2, graph convolution on GO. A two-layer graph convolution over the union of nine gene-gene graphs on the propagated Gene Ontology multi-hot (2,362 terms over 5,501 annotated genes), which reads 0.917 [0.870, 0.955] as a five-fold ensemble. It is inductive in the regime key's sense: a gene's own annotation and graph position, no call of any other gene.

Their average. Each family's logits standardized and weighted equally with nothing fitted. The best of the 10 two-family averages on file is 0.955 [0.922, 0.982] (GO convolution, five folds (400 epochs) with + GPR redundancy counts, pathway + ProtT5, five folds (round 80)); the average was fixed in advance before the 10 repeated splits were read, where it beats the layer-free form by +0.048 (10 of 10) and graph convolution on GO by +0.026 (9 of 10).

Comparators on the same genes. Flux Cone Learning (Merzbacher 2025) scores 0.742 on the 195 genes; on its own released held-out set of 223 it reads 0.731, and its published five-fold mean of 0.780 was recomputed at 0.777 from the released notebook, so the comparison is against a reproduced baseline. FBA with yeast-GEM 9.0.2 and nothing fitted scores 0.640 [0.519, 0.755]. A DCell-style model over 2,362 GO terms, trained under the same fold protocol, reads 0.848 [0.769, 0.912] as a fold ensemble (fold mean 0.827).

The genome-wide limit. On the 4,674 genes with no GEM reaction the best arm on file is graph convolution on ProtT5 + FBA ratios at 0.759; the transformer with the solve reads 0.570, graph convolution on ProtT5 over the union 0.755, graph convolution on the physical graph's degree and clustering 0.759, and the FBA ratio 0.500 (a constant). Every genome-wide arm is in 7.

4 What is established and what remains ✗

Established. The held-out score against two comparators on the same genes, with the margin held on 10 repeated splits; the module form (an offline solve read as data through an affine) and that it carries across both families and inside one medium; the genome-wide limit of the metabolic columns and the gene graphs' coverage of the rest; the hard genes and their shared annotation (in `028-gene-essentiality`).

Not established. The main-text composition at Nature size from the same result files (R1). Seeds and intervals on every bar of that composition drawn from one script, not from the two summary sheets (R1). Whether the best average holds on a split that is not the released one when the choice of average is made on the repeated splits themselves rather than fixed in advance (R2). The link to the solid-growth model: the 030 holdout arms serve 698 essentiality singles as a second validation loader, and no AUROC from them is on file (R3). Whether the essentiality call carries to a second host with a genome-scale model and no screen, which is the generality claim the figure cannot yet make (R4).

Corrections recorded. The 028 summary quotes the best single model at 0.934; the fold-ensemble file this document reads gives 0.930 for + GPR redundancy counts, pathway + ProtT5, five folds (round 80), and the 0.934 is the one-seed fold ensemble of the four-media form in the simplification ladder (0.934). Both are on file; the figure uses the fold-ensemble table, and the caption says which.

5 The decision rules, fixed now ✗

What a panel needs to enter the paper. Every drawn value in a committed result file (●); every learned bar a fold ensemble or a seed mean with its interval or spread drawn; the regime of every bar named in the key; the comparators on the same genes; and the claim the panel carries written in one sentence in this document before the panel is composed.

The claims and their rules. The score claim stands if the best average's gene bootstrap interval excludes the Flux Cone Learning point and the average beats the better single model on at least eight of the 10 repeated splits; on file it is 9 of 10. The module claim stands if the single-medium form beats no metabolism on at least eight of 10 splits and matches the four-media form within 0.01; on file 10 of 10 and +0.002. The coverage claim stands as a statement of the data (which genes carry a reaction) and needs no rule. The generality claim is not made.

Order of work. R1 first, since it is composition from files that exist. R3 is one readout script over 030 runs that are queued already. R2 is CPU. R4 is a campaign and is not this figure's.

6 The experiment registry ✗

Shared protocol: the released held-out genes never in a loss; the epoch chosen on the validation folds; AUROC with a gene bootstrap of 2,000 draws; the repeated splits at three seeds each. Every entry is ❍ until its result file exists.

Table 1: The pre-registered items of this gate.

id	item	
R1	**The main-text composition.** One script in `experiments/028-gene-essentiality/scripts/` draws the six panels of 2 at Nature's full-page box from the same result files as the summary sheets, with the families explainer re-exported from its draw.io source; every number in its caption a macro of this document. Decides: the figure's entry.	❍
R2	**The average chosen on the repeated splits.** Nested: choose the pair to average on nine of the ten repeated splits and read it on the tenth, around the ring. Decides: whether the fixed-in-advance margin survives a data-driven choice.	❍
R3	**Essentiality from the solid-growth model.** The AUROC of the 030 holdout arms (`fit_000`, `embfit_001`) on their 698 served singles, read from the run registry, beside the 028 families on the same genes where they overlap. Decides: the sentence linking this figure to Figure 2.	❍
R4	**A second host.** The layer-free solve and the GO convolution on a bacterial host with a genome-scale model and an essentiality screen, trained on yeast and read there, and trained there and read there. Decides: the generality claim, which this figure does not make until it reads.	❍

7 Every ensemble and every genome-wide arm ✗

Every fold ensemble and post hoc average on the 195 released held-out genes, best first, with its gene-bootstrap interval; then every arm read on the 4,674 genome-wide genes with no GEM reaction, pooled over its cells.

Table 2: The 23 fold ensembles and averages on the released held-out genes. Rows written by `essentiality_gate_numbers.py`.

models	kind	cells	AUROC	95 % interval
GO convolution, five folds (400 epochs) with + GPR redundancy counts, pathway + ProtT5, five folds (round 80)	average	10	0.955	[0.922, 0.982]
GO convolution, five folds (400 epochs) with cached solve + flux head, pathway + ProtT5, five folds (round 27)	average	10	0.954	[0.921, 0.978]
GO convolution, five folds (400 epochs) with layer-free log1p vector + affine, pathway + ProtT5, five folds (round 34)	average	10	0.953	[0.920, 0.979]
GO convolution, five folds (400 epochs) with YPD ratio + vector + counts, pathway + ProtT5, five folds (rounds 84 to 98)	average	10	0.953	[0.915, 0.980]
GO convolution, five folds (400 epochs) with layer-free log1p, interior-point store, five folds (round 37)	average	10	0.951	[0.916, 0.978]
GO convolution, five folds (400 epochs) with flux head, pathway + ProtT5, five folds (round 25)	average	10	0.949	[0.912, 0.977]
GO convolution, five folds (400 epochs) with layer-free log1p, PDLP store, five folds (round 39)	average	10	0.949	[0.912, 0.978]
GO convolution, five folds (400 epochs) with gene-rule counts, no solve, five folds (round 82)	average	10	0.948	[0.908, 0.979]
GO convolution, five folds (400 epochs) with indicator + ProtT5 + affine, five folds (round 12) with cached solve + flux head, pathway + ProtT5, five folds (round 27)	average	15	0.947	[0.912, 0.976]
GO convolution, five folds (400 epochs) with indicator + ProtT5 + affine, five folds (round 12) with flux head, pathway + ProtT5, five folds (round 25)	average	15	0.945	[0.907, 0.974]
GO convolution, five folds (400 epochs) with indicator + ProtT5 + affine, five folds (round 12)	average	10	0.944	[0.903, 0.975]
GO convolution with the module as an affine, five folds (job 2695) with YPD ratio + vector + counts, pathway + ProtT5, five folds (rounds 84 to 98)	average	10	0.941	[0.894, 0.976]
+ GPR redundancy counts, pathway + ProtT5, five folds (round 80)	single family	5	0.930	[0.886, 0.966]
GO convolution with the module as an affine, five folds (job 2695)	single family	5	0.930	[0.877, 0.969]
YPD ratio + vector + counts, pathway + ProtT5, five folds (rounds 84 to 98)	single family	5	0.929	[0.885, 0.966]
cached solve + flux head, pathway + ProtT5, five folds (round 27)	single family	5	0.922	[0.877, 0.959]
layer-free log1p vector + affine, pathway + ProtT5, five folds (round 34)	single family	5	0.920	[0.873, 0.957]
GO convolution, five folds (400 epochs)	single family	5	0.917	[0.870, 0.955]
layer-free log1p, PDLP store, five folds (round 39)	single family	5	0.915	[0.870, 0.954]
layer-free log1p, interior-point store, five folds (round 37)	single family	5	0.913	[0.868, 0.953]
gene-rule counts, no solve, five folds (round 82)	single family	5	0.909	[0.858, 0.954]
indicator + ProtT5 + affine, five folds (round 12)	single family	5	0.897	[0.839, 0.944]
flux head, pathway + ProtT5, five folds (round 25)	single family	5	0.897	[0.841, 0.944]

Table 3: Every arm on the genome-wide genes with no GEM reaction, pooled AUROC over cells with a gene bootstrap. Rows written by `essentiality_gate_numbers.py`.

arm	kind	cells	AUROC	95 % interval
graph convolution on ProtT5 + FBA ratios	GCN	3	0.759	[0.740, 0.777]
graph convolution, physical graph, degree and clustering	GCN	3	0.759	[0.740, 0.777]
graph convolution, union of nine, ProtT5	GCN	3	0.755	[0.737, 0.773]
graph convolution on ProtT5 alone, 400 epochs	GCN	3	0.749	[0.731, 0.768]
graph convolution on the three inputs, 400 epochs	GCN	3	0.744	[0.725, 0.762]
graph attention, union of nine, degree and clustering	GATv2	3	0.717	[0.699, 0.735]
graph attention, physical graph, degree and clustering	GATv2	3	0.708	[0.689, 0.728]
graph convolution on GO alone, 400 epochs	GCN	3	0.703	[0.685, 0.722]
graph convolution on ProtT5 + GO + FBA ratios, 400 epochs	GCN	3	0.702	[0.682, 0.721]
graph convolution on GO + FBA ratios, 400 epochs	GCN	3	0.700	[0.680, 0.719]
graph convolution, physical graph, all-ones input	GCN	3	0.686	[0.668, 0.704]
graph convolution, union of nine, degree and clustering	GCN	3	0.663	[0.644, 0.683]
graph convolution, union of nine, all-ones input	GCN	3	0.643	[0.624, 0.661]
layer-free + physical degree and clustering in the affine (round 76)	transformer	6	0.593	[0.573, 0.613]
pathway multi-hot + ProtT5 + FBA affine (round 22 recipe)	transformer	6	0.572	[0.552, 0.591]
layer-free log1p vector + affine, pathway + ProtT5 (round 33 form)	transformer	6	0.570	[0.550, 0.590]
layer-free, store over every gene (wild-type rows; round 76 control)	transformer	6	0.570	[0.550, 0.590]
cached solve + flux head, pathway + ProtT5 (round 27 form)	transformer	6	0.551	[0.530, 0.570]
graph convolution, physical graph, ProtT5	GCN	3	0.544	[0.523, 0.566]
MLP on ProtT5 alone	MLP	3	0.534	[0.514, 0.554]
MLP on the three inputs	MLP	3	0.509	[0.490, 0.528]
FBA knockout ratio on YPD (0 of 4674 genes carry a reaction)	solver	1	0.500	[0.500, 0.500]
MLP on ProtT5 + FBA ratios	MLP	3	0.481	[0.462, 0.502]